Queen Mary
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EBU5376
Digital Signal Processing

Infinite Impulse Response (IIR) Filters

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Outline

- Definition of Infinite Impulse Response (IIR) Filter
- Bilinear Transform
  - Frequency Warping
- IIR Example: Butterworth Filter

There will be MATLAB exercises throughout this lecture.

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Infinite Impulse Response (FIR) Filter

- **Infinite Impulse Response (IIR) system**
  - If the impulse response $h[n]$ is an infinite sequence, the system is called an Infinite Impulse Response (IIR) system.
  - This happens when the difference equation contains past outputs $y[n-1], y[n-2] \ldots,$

Difference equation of IIR system:

$$y[n] = -\sum_{i=1}^{N} a_i y[n-i] + \sum_{j=0}^{M} b_j x[n-j]$$

Infinite Impulse Response (IIR) Filter

- Consider the general transfer function formula

$$H(z) = \frac{Y(z)}{X(z)} = \frac{\sum_{k=0}^{M} b_k z^{-k}}{\sum_{k=0}^{N} a_k z^{-k}}, \qquad a_0 = 1$$

- It is an **IIR filter** if and only if $a_k \neq 0, \text{ for some } k > 1$.

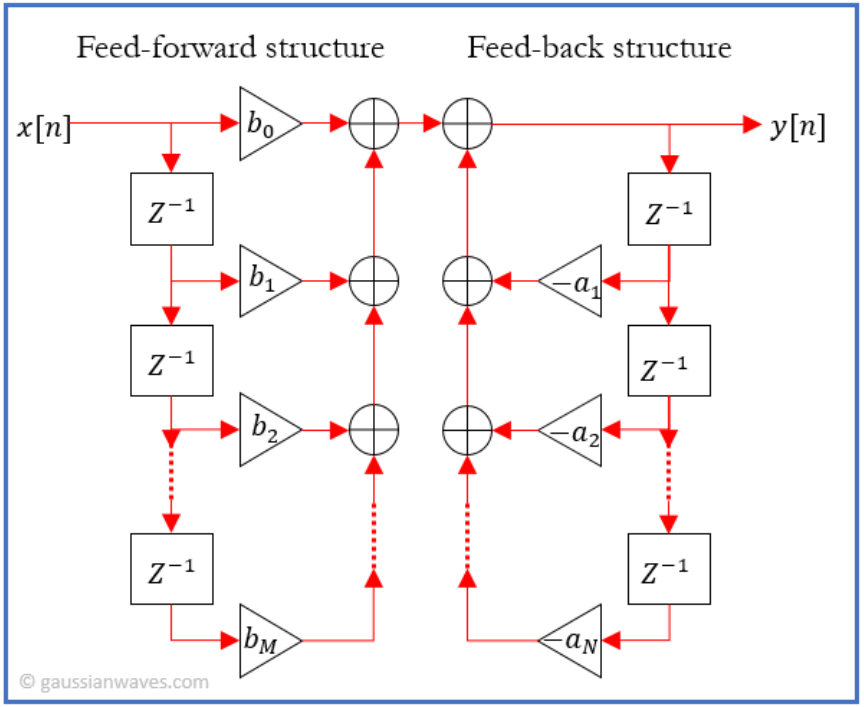
Transfer function of IIR filter:

$$H(z) = \frac{Y(z)}{X(z)} = \frac{\sum_{k=0}^{M} b_k z^{-k}}{\sum_{k=0}^{N} a_k z^{-k}}$$

- The transfer function <u>contains both poles and zeros</u>
- The difference equation is recursive, i.e., this is a <u>recursive filter</u>

Infinite Impulse Response (IIR) Filter

- Typical Block Diagram of IIR filters:



- Consists of both a feed-forward structure and a feed-back structure.
- Past outputs are involved.
- As n increases, the feedback structure becomes a set of infinite loops

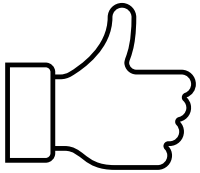
Infinite Impulse Response (IIR) Filter

- IIR filters have **infinite impulse responses**
- This is because when an impulse is applied to the input of the filter, the filter's output is a weighted sum of its past outputs and input samples, which are fed back into the filter.
- The feedback causes the output to depend on previous output values, and this dependence can continue indefinitely, leading to an infinite impulse response.

Infinite Impulse Response (IIR) Filter

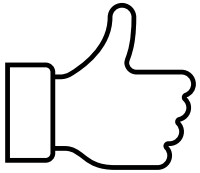
- IIR filters have **both poles and zeros**
- In contrast to FIR, which only has zeros, IIR filters must have as least one pole.
- Since the stability of a filter is determined by the location of the pole, IIR filter may be unstable.
- Hence, *careful attention must be paid* to the design and implementation of IIR filter.
- The existence of poles also leads to non-linear phase response

Infinite Impulse Response (IIR) Filter



- IIR filters also have **advantages**
- The *poles* of an IIR filter enable it to achieve greater attenuation of higher frequency components in the input signal
- It allows for the design of filters with sharper frequency roll-offs than FIR filters with the same number of filter coefficients
- This is important in applications such as audio and speech processing

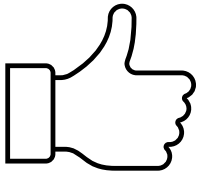
Infinite Impulse Response (IIR) Filter



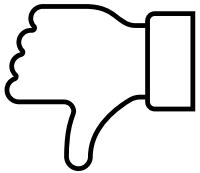
- IIR filters also have **advantages**
- To ensure the stability, IIR filters often have fewer filter coefficients than FIR filters. Hence, IIR is more computationally efficiency than FIR
- Hence, IIR filters are more suitable for real-time processing, where low latency and fast response times are essential
- Also, fewer coefficients mean IIR filters are more suitable for applications with limited computational resources, such as mobile devices

Infinite Impulse Response (IIR) Filter

- Advantages of IIR
  1. Low computational complexity
  2. Cheap and easy to implement
  3. Sharper frequency roll-offs



- Disadvantages of IIR
  1. Can be unstable
  2. Non-linear phase response



Outline

- Definition of Infinite Impulse Response (IIR) Filter
- Bilinear Transform
  - Frequency Warping
- IIR Example: Butterworth Filter

There will be MATLAB exercises throughout this lecture.

Design of IIR Filter

- The most common design method for digital IIR filters is based on designing an analog IIR filter and then converting it to an equivalent digital filter
- There are two common methods of transformation:
  1. **Bilinear transform** — Matches transfer function
  2. Impulse Invariant method — Matches impulse response
- We will learn the method of bilinear transform because it *provides a better balance between the distortions of frequency and time domain*

Definition of Bilinear Transform

- The **bilinear transform** is a common method in DSP and control theory to transform continuous-time systems to discrete-time systems, i.e., it performs discrete-time approximation
- It is defined by the substitution

$$s = \frac{2}{T}\frac{1 - z^{-1}}{1 + z^{-1}}$$

  - where s is the operator of Laplace transform and z is the operator of z-transform
  - T is the sampling interval in discrete-time

Design of IIR Filter – Bilinear Transform (BLT)

- We start with the general transfer function of an IIR analog filter

$$H(s) = K\frac{(s-q_1)(s-q_2)\ldots}{(s-p_1)(s-p_2)\ldots}$$

  - For analog filters, the transfer function is the Laplace transform of its impulse response
  - Hence, the parameter s is the Laplace operator $s = \sigma + j\omega$
  - $q_1, q_2, \ldots$ are the <u>zeros</u> of this analog filter and $p_1, p_2, \ldots$ are the <u>poles</u> of this analog filter

Design of IIR Filter – Bilinear Transform (BLT)

- We start with the general transfer function of an IIR analog filter

$$H(s) = K\frac{(s-q_1)(s-q_2)\ldots}{(s-p_1)(s-p_2)\ldots}$$

- To apply bilinear transform, we substitute $s = \frac{2}{T}\frac{1-z^{-1}}{1+z^{-1}}$ to $H(s)$:

$$\boxed{H(z) = H(s)\Big|_{s=\frac{2}{T}\frac{1-z^{-1}}{1+z^{-1}}}}$$

- This is the *transfer function mapping* by bilinear transform

Design of IIR Filter – Bilinear Transform (BLT)

- Example 1: Using bilinear transform to convert the following analog filter into a digital filter with a sampling interval of 1 second

$$H(s) = \frac{2}{(s+1)(s+2)}$$

Design of IIR Filter – Bilinear Transform (BLT)

- Example 1: Using bilinear transform to convert the following analog filter into a digital filter with a sampling interval of 1 second

$$H(s) = \frac{2}{(s+1)(s+2)}$$

$$H(z) = H(s)\Big|_{s=\frac{2}{T}\frac{1-z^{-1}}{1+z^{-1}}}$$

$$= \frac{2}{\left(2\frac{1-z^{-1}}{1+z^{-1}}+1\right)\left(2\frac{1-z^{-1}}{1+z^{-1}}+2\right)} = \frac{2}{\left(\frac{4(3-z^{-1})}{(1+z^{-1})^2}\right)} = \frac{(1+z^{-1})^2}{6-2z^{-1}}$$

$$= \frac{\frac{1}{6}(1+z^{-1})^2}{1-\frac{1}{3}z^{-1}}$$

The coefficient of $y[n]$ must be 1

Design of IIR Filter – Bilinear Transform (BLT)

- Example 2: Using bilinear transform to convert the following analog filter into a digital filter with a sampling interval of 1 second

$$H(s) = \frac{s+1}{s^2+5s+6}$$

Design of IIR Filter – Bilinear Transform (BLT)

- Example 2: Using bilinear transform to convert the following analog filter into a digital filter with a sampling interval of 1 second

$$H(s) = \frac{s+1}{s^2+5s+6}$$

$$H(z) = H(s)\Big|_{s=\frac{2}{T}\frac{1-z^{-1}}{1+z^{-1}}}$$

$$= \frac{2\frac{1-z^{-1}}{1+z^{-1}}+1}{\left(2\frac{1-z^{-1}}{1+z^{-1}}\right)^2+5\left(2\frac{1-z^{-1}}{1+z^{-1}}\right)+6} = \frac{\frac{3-z^{-1}}{1+z^{-1}}}{\frac{(5+z^{-1})\cdot 4}{(1+z^{-1})^2}} = \frac{(3-z^{-1})(1+z^{-1})}{4(5+z^{-1})}$$

$$= \frac{3+2z^{-1}-z^{-2}}{20+4z^{-1}} = \frac{0.15+0.1z^{-1}-0.05z^{-2}}{1+0.2z^{-1}}$$

Design of IIR Filter – Bilinear Transform (BLT)

- Below is the MATLAB function for performing the BLT (MATLAB link):
  - [zd,pd,kd] = bilinear(z,p,k,fs)
- The inputs are
  - ‘z’ → the zeros of the analog filter (a column vector)
  - ‘p’ → the poles of the analog filter (a column vector)
  - ‘k’ → gain of the analog filter (a constant)
  - ‘fs’ → sampling frequency (a constant)
- The outputs are
  - ‘zd’ → the zeros of the digital filter (a column vector)
  - ‘pd’ → the poles of the digital filter (a column vector)
  - ‘kd’ → gain of the digital filter (a constant)

Design of IIR Filter – Bilinear Transform (BLT)

- MATLAB Example 1: Using MATLAB, apply BLT to the following analog filter with sampling interval $T = 1$ second

$$H(s) = \frac{2}{(s+1)(s+2)}$$

Inputs:

```
z=[;] % zeros of analog filter
```

```
z =

     []
```

```
p=[-1;-2] % poles of analog filter
```

```
p = 2×1
    -1
    -2
```

```
k=2     % gain of analog filter
```

```
k = 2
```

```
fs=1/1  % sampling frequency
```

```
fs = 1
```

Function:

```
[zd,pd,kd]=bilinear(z,p,k,fs)
```

Design of IIR Filter – Bilinear Transform (BLT)

- MATLAB Example 1: Using MATLAB, apply BLT to the following analog filter with sampling interval $T = 1$ second

$$H(s) = \frac{2}{(s+1)(s+2)}$$

Outputs:

```
[zd,pd,kd]=bilinear(z,p,k,fs)
```

```
zd = 2×1
    -1
    -1

pd = 2×1
    0.3333
         0

kd = 0.1667
```

Zeros of the digital filter

Poles of the digital filter

Gain of the digital filter

$$H(z) = \frac{\frac{1}{6}(1+z^{-1})^2}{1-\frac{1}{3}z^{-1}}$$

Same as our answer!

Design of IIR Filter – Bilinear Transform (BLT)

- Below is *an alternative way* of using the previous MATLAB function for performing the BLT (MATLAB link):
  - [numd,dend] = bilinear(num,den,fs)
- The inputs are
  - 'num' → numerator coeffs. of the analog transfer function (a row vector)
  - 'den' → denominator coeffs. of the analog transfer function (a row vector)
  - 'fs' → sampling frequency (a constant)
- The outputs are
  - 'numd' → numerator coeffs. of the digital transfer function (a row vector)
  - 'dend' → denominator coeffs. of the digital transfer function (a row vector)

Design of IIR Filter – Bilinear Transform (BLT)

- MATLAB Example 2: Using MATLAB, apply BLT to the following analog filter with sampling interval $T = 1$

$$H(s) = \frac{s+1}{s^2+5s+6}$$

Inputs:

```
num=[0,1,1] % the numerator of digital transfer function
```

```
num = 1×3
     0     1     1
```

```
fs=1/1      % the sampling frequency
```

```
fs = 1
```

```
den=[1,5,6] % the denominator of digital transfer function
```

```
den = 1×3
     1     5     6
```

- ✓ 'num' and 'den' should have the same dimension
- ✓ Start with the coeff. of the **highest order**

Function:

```
[numd,dend]=bilinear(num,den,fs)
```

Design of IIR Filter – Bilinear Transform (BLT)

- MATLAB Example 2: Using MATLAB, apply BLT to the following analog filter with sampling interval $T = 1$

$$H(s) = \frac{s+1}{s^2+5s+6}$$

Outputs:

```
[numd,dend]=bilinear(num,den,fs)
```

Numerator coeffs. of the digital filter

```
numd = 1×3
    0.1500    0.1000   -0.0500
```

$$H(z) = \frac{0.15 + 0.1z^{-1} - 0.05z^{-2}}{1 + 0.2z^{-1}}$$

```
dend = 1×3
    1.0000    0.2000   -0.0000
```

Denominator coeffs. of the digital filter

Same as our answer!

Design of IIR Filter – Bilinear Transform (BLT)

- Conclusion on using MATLAB for BLT
  - If the analog **transfer function** has the pole-zero form:

$$H(s) = K\frac{(s-q_1)(s-q_2)\ldots}{(s-p_1)(s-p_2)\ldots}$$

Use [zd,pd,kd] = bilinear(z,p,k,fs)

  - If the analog **transfer function** has the polynomial form:

$$H(s) = \frac{\alpha_0 + \alpha_1 s + \alpha_2 s^2 + \cdots}{1 + \beta_1 s + \beta_2 s^2 + \cdots}$$

Use [numd,dend] = bilinear(num,den,fs)

Design of IIR Filter – Bilinear Transform (BLT)

- Previously we obtained the transfer function mapping by BLT
- Next, we examine the frequency response mapping under BLT
- We denote the frequency of the analog filter by Ω and the frequency of the digital filter by ω
- The relationship between them is given by

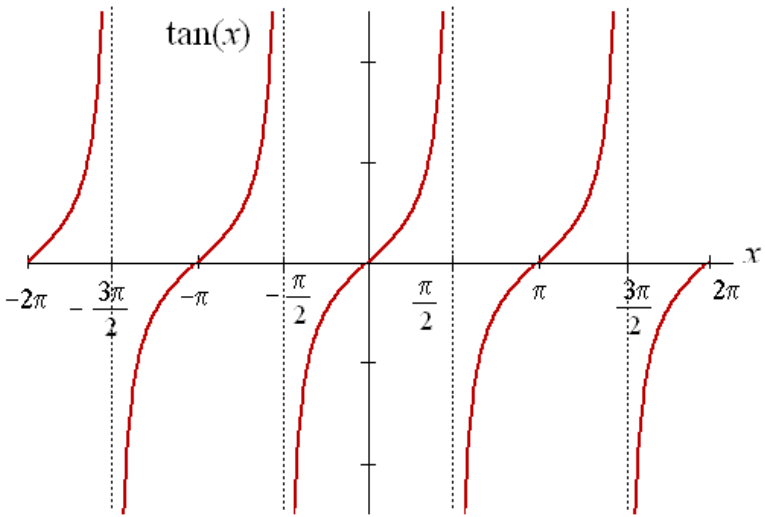
$$\Omega = \frac{2}{T}\tan\frac{\omega T}{2}$$

  where T is the sampling interval
  - The derivation to this equation can be found in the Appendix
- This non-linear relationship is known as the **frequency warping effect**, which occurs when converting between analog and digital frequencies

Design of IIR Filter – Bilinear Transform (BLT)

$$\Omega = \frac{2}{T}\tan\frac{\omega T}{2}$$

- <u>Frequency Warping</u>
  - This is because Ω ranges from $-\infty$ to ∞, but ω ranges from $-\pi$ to π
  - For small values of ω, $\tan\frac{\omega T}{2} \approx \frac{\omega T}{2}$.
    Thus, this relationship is linear $\rightarrow \Omega \approx \omega$
  - The frequency distortion occurs as the frequency increases
  - Hence, when applying BLT, we need to pre-warp the analog frequency to prevent this distortion



Design of IIR Filter – Bilinear Transform (BLT)

- <u>Frequency Warping</u> $\Omega = \frac{2}{T}\tan\frac{\omega T}{2} \Leftrightarrow \omega = \frac{2}{T}\tan^{-1}\frac{\Omega T}{2}$
  - Why is frequency warping a bad thing?
    - Suppose we want a digital filter with a cut-off frequency at $\omega_c = 1$ rad/sec.
    - We picked an analog prototype that satisfy this requirement ($\Omega_c = 1$), and convert it to a digital filter via BLT with interval $T = 1$
    - The resulting digital filter will have a cut-off frequency of

$$\omega_c = 2\tan^{-1}\frac{\Omega_c}{2} = 0.9273$$

    - The digital filter does not fit the requirement!

Design of IIR Filter – Bilinear Transform (BLT)

- Frequency Pre-Warping

$$\Omega = \frac{2}{T}\tan\frac{\omega T}{2} \Leftrightarrow \omega = \frac{2}{T}\tan^{-1}\frac{\Omega T}{2}$$

  - To remedy this effect, we pre-warp the analog frequency
    - Suppose we want a digital filter with a cut-off frequency at $\omega_c = 1$ rad/sec.
    - We picked an analog prototype that satisfy a pre-warped requirement ($\Omega_c = \frac{2}{T}\tan\frac{T}{2}$), and convert it to a digital fiter via BLT with interval T
    - The resulting digital filter will have a cut-off frequency of $\omega_c = \frac{2}{T}\tan^{-1}\frac{\Omega_c T}{2} = 1$
    - The digital filter successfully fit the requirement!

Outline

- Definition of Infinite Impulse Response (IIR) Filter
- Bilinear Transform
  - Frequency Warping
- IIR Example: Butterworth Filter

There will be MATLAB exercises throughout this lecture.

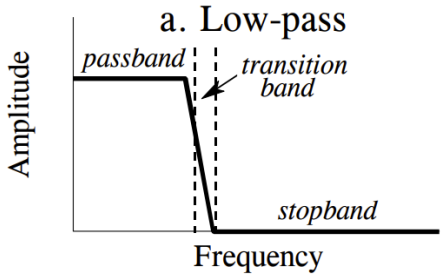
IIR Filter Example – Butterworth Filter

- The **Butterworth filter** (also called **maximally flat filter**) is a low-pass IIR filter that has a flat amplitude response in the passband and a gradual roll-off in the stopband

- It is named after the British engineer and physicist Stephen Butterworth, who first described the filter in 1930.

- An analog Butterworth filter is defined with an amplitude response of

$$|H_a(\omega)| = \frac{1}{\sqrt{1 + \left(\frac{\omega}{\omega_c}\right)^{2n}}}$$



- ✓ n is the order of the filter
- ✓ ω_c is the cut-off frequency

IIR Filter Example – Butterworth Filter

$$|H_a(\omega)| = \frac{1}{\sqrt{1 + \left(\frac{\omega}{\omega_c}\right)^{2n}}}$$

- The Butterworth filter is characterized by its order n, which determines the steepness of the roll-off in the stopband.
- Higher order Butterworth filters have a steeper roll-off, but also exhibit *more phase distortion* and *becomes unstable*

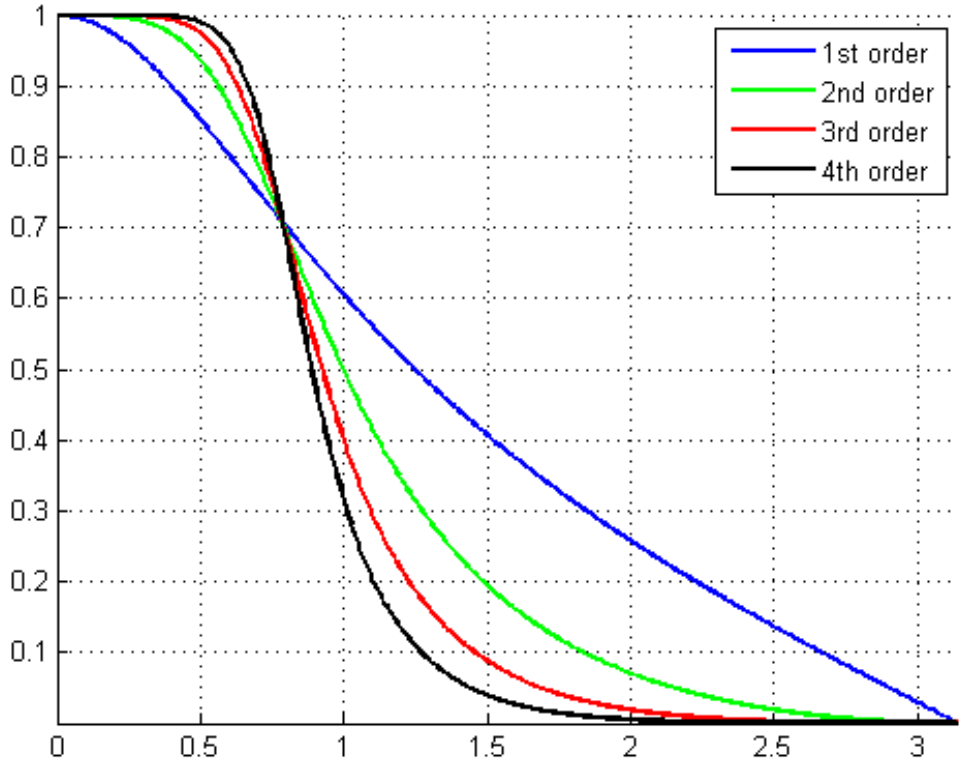


Fig: Amplitude response of Butterworth filter with different order

IIR Filter Example – Butterworth Filter

- **Transfer function** of typical Butterworth filters ($\omega_c = 1$) at each order:
  - Order $n = 1$

$$H(s) = \frac{1}{s+1}$$

  - Order $n = 2$

$$H(s) = \frac{1}{s^2 + \sqrt{2}s + 1}$$

  - Order $n = 3$

$$H(s) = \frac{1}{(s+1)(s^2+s+1)}$$

IIR Filter Example – Butterworth Filter

- **Transfer function** of general Butterworth filters at each order:
  - Order $n = 1$

$$H(s) = \frac{1}{\frac{s}{\omega_c} + 1}$$

  - Order $n = 2$

$$H(s) = \frac{1}{\left(\frac{s}{\omega_c}\right)^2 + \sqrt{2}\left(\frac{s}{\omega_c}\right) + 1}$$

  - Order $n = 3$

...

IIR Filter Example – Butterworth Filter

- Example 1: Consider an IIR filter with a cut-off frequency of $\omega_c = 1$ rad/s.
  1. Construct this filter using an order-1 Butterworth prototype ($n = 1$) under sampling interval $T = 1$ without pre-warping.
  2. Check your BLT answer using MATLAB.

IIR Filter Example – Butterworth Filter

- Example 1: Consider an IIR filter with a cut-off frequency of $\omega_c = 1$ rad/s.
  1. Construct this filter using an order-1 Butterworth prototype ($n = 1$) under sampling interval $T = 1$ without pre-warping.
  2. Check your BLT answer using MATLAB.

Order 1 Butterworth with $\omega_c = 1$ is

$$H(s) = \frac{1}{s+1}$$

Apply BLT to $H(s)$ via the following equation:

$$H(z) = H(s)\Big|_{s=\frac{2}{T}\frac{1-z^{-1}}{1+z^{-1}}}$$

IIR Filter Example – Butterworth Filter

- Example 1: Consider an IIR filter with a cut-off frequency of $\omega_c = 1$ rad/s.
  1. Construct this filter using an order-1 Butterworth prototype ($n = 1$) under sampling interval $T = 1$ without pre-warping.
  2. Check your BLT answer using MATLAB.

Then, evaluate $H(z) = H(s)\big|_{s=\frac{2}{T}\frac{1-z^{-1}}{1+z^{-1}}}$

$$= \frac{1}{2\dfrac{1-z^{-1}}{1+z^{-1}}+1} = \frac{1+z^{-1}}{3-z^{-1}} = \frac{\frac{1}{3}+\frac{1}{3}z^{-1}}{1-\frac{1}{3}z^{-1}}$$

$$= \frac{0.333+0.333z^{-1}}{1-0.333z^{-1}}$$

IIR Filter Example – Butterworth Filter

- Example 1: Consider an IIR filter with a cut-off frequency of $\omega_c = 1$ rad/s.
  1. Construct this filter using an order-1 Butterworth prototype ($n = 1$) under sampling interval $T = 1$ without pre-warping.
  2. Check your BLT answer using MATLAB.

MATLAB command for BLT:

```
[numd,dend]=bilinear(num,den,fs)
```

We need to define the inputs:

1. num
2. den
3. fs

```
num=[0,1];  % -> 1
den=[1,1];  % -> s+1
fs=1;
```

IIR Filter Example – Butterworth Filter

- Example 1: Consider an IIR filter with a cut-off frequency of $\omega_c = 1$ rad/s.
  1. Construct this filter using an order-1 Butterworth prototype ($n = 1$) under sampling interval $T = 1$ without pre-warping.
  2. Check your BLT answer using MATLAB.

```
num=[0,1];  % -> 1
den=[1,1];  % -> s+1
fs=1;
[numd,dend]=bilinear(num,den,fs)
```

```
numd = 1×2
      0.3333    0.3333

dend = 1×2
      1.0000   -0.3333
```

Same answer as in part 1

$$0.333 + 0.333z^{-1}$$

$$1 - 0.333z^{-1}$$

IIR Filter Example – Butterworth Filter

- Example 2: Consider an IIR filter with a cut-off frequency of $\omega_c = 1$ rad/s.
  1. Construct this filter using an order-1 Butterworth prototype ($n = 1$) under sampling interval $T = 1$ with pre-warping.
  2. Check your BLT answer using MATLAB

IIR Filter Example – Butterworth Filter

- Example 2: Consider an IIR filter with a cut-off frequency of $\omega_c = 1$ rad/s.
  1. Construct this filter using an order-1 Butterworth prototype ($n = 1$) under sampling interval $T = 1$ with pre-warping.
  2. Check your BLT answer using MATLAB

Three steps:

Step 1: Pre-warp the analog cut-off frequency

Step 2: Formulate the order-1 Butterworth prototype

Step 3: Perform BLT

IIR Filter Example – Butterworth Filter

- Example 2: Consider an IIR filter with a cut-off frequency of $\omega_c = 1$ rad/s.
  1. Construct this filter using an order-1 Butterworth prototype ($n = 1$) under sampling interval $T = 1$ with pre-warping.
  2. Check your BLT answer using MATLAB

Pre-warp the cut-off frequency by:

$$\Omega_c = \frac{2}{T}\tan\frac{\omega_c T}{2}$$
$$= 2\tan\frac{1}{2}$$
$$= 1.0926 \text{ rad/s}$$

Equation of order-1 Butterworth filter:

$$H(s) = \frac{1}{\frac{s}{\omega_c} + 1}$$
$$= \frac{1.0926}{s + 1.0926}$$

IIR Filter Example – Butterworth Filter

- Example 2: Consider an IIR filter with a cut-off frequency of $\omega_c = 1$ rad/s.
  1. Construct this filter using an order-1 Butterworth prototype ($n = 1$) under sampling interval $T = 1$ with pre-warping.
  2. Check your BLT answer using MATLAB

Perform BLT to $H(s) = \frac{1.0926}{s+1.0926}$

$$H(z) = H(s)\Big|_{s=\frac{2}{T}\frac{1-z^{-1}}{1+z^{-1}}}$$

$$= \frac{1.0926}{2\frac{1-z^{-1}}{1+z^{-1}} + 1.0926} = \frac{1.0926 + 1.0926z^{-1}}{2 - 2z^{-1} + 1.0926 + 1.0926z^{-1}}$$

$$= \frac{1.0926 + 1.0926z^{-1}}{3.0926 - 1.0926z^{-1}} = \frac{0.3533 + 0.3533z^{-1}}{1 - 0.2934z^{-1}}$$

Pre-warping changes the digital transfer function

IIR Filter Example – Butterworth Filter

- Example 2: Consider an IIR filter with a cut-off frequency of $\omega_c = 1$ rad/s.
  1. Construct this filter using an order-1 Butterworth prototype ($n = 1$) under sampling interval $T = 1$ with pre-warping.
  2. Check your BLT answer using MATLAB

MATLAB command for BLT:

```
[numd,dend]=bilinear(num,den,fs)
```

We need to define the inputs:

1. num
2. den
3. fs

```
num=[0,1.0926];   % -> 1.0926
den=[1,1.0926];   % -> s+1.0926
fs=1;
```

IIR Filter Example – Butterworth Filter

- Example 2: Consider an IIR filter with a cut-off frequency of $\omega_c = 1$ rad/s.
  1. Construct this filter using an order-1 Butterworth prototype ($n = 1$) under sampling interval $T = 1$ with pre-warping.
  2. Check your BLT answer using MATLAB

```
num=[0,1.0926];   % -> 1.0926
den=[1,1.0926];   % -> s+1.0926
fs=1;
[numd,dend]=bilinear(num,den,fs)
```

```
numd = 1×2
      0.3533    0.3533
```

$$0.3533 + 0.3533z^{-1}$$

```
dend = 1×2
      1.0000   -0.2934
```

$$1 - 0.2934z^{-1}$$

Same answer as in part 1

IIR Filter Example – Butterworth Filter

- Create Butterworth Prototype ($\omega_c = 1$) using MATLAB:

```
[z,p,k] = buttap(n)
```

- Input:
  - ‘n’ → the order of the Butterworth prototype
- Output:
  - ‘z’ → zero(s) of the Butterworth prototype ([ ], empty array)
  - ‘p’ → pole(s) of the Butterworth prototype (column vector)
  - ‘k’ → gain of the Butterworth prototype (constant)

$$H(s) = K\frac{(s-q_1)(s-q_2)\ldots}{(s-p_1)(s-p_2)\ldots}$$

IIR Filter Example – Butterworth Filter

- Example: Use MATLAB to create an order-2 Butterworth ($\omega_c = 1$) prototype. Use the outputs to write out the transfer function of this Butterworth filter.

```
[z,p,k] = buttap(2)
```

```
z =

     []

p = 2×1 complex

   -0.7071 + 0.7071i
   -0.7071 - 0.7071i

k = 1
```

$$H(s) = K\frac{(s-q_1)(s-q_2)\ldots}{(s-p_1)(s-p_2)\ldots}$$

$$\frac{\sqrt{2}}{2} \approx 0.7071$$

$$H(s) = \frac{1}{\left(s-\left(-\frac{\sqrt{2}}{2}+\frac{\sqrt{2}}{2}j\right)\right)\left(s-\left(-\frac{\sqrt{2}}{2}-\frac{\sqrt{2}}{2}j\right)\right)} = \frac{1}{s^2+\sqrt{2}s+1}$$

Lecture summary

- IIR filters are defined with a difference equation that contains at least one past output
- In terms of the general equation

$$H(z) = \frac{Y(z)}{X(z)} = \frac{\sum_{k=0}^{M} b_k z^{-k}}{\sum_{k=0}^{N} a_k z^{-k}},$$

We define it to be an IIR filter if $a_k \neq 0, for\ some\ k > 1$.

- IIR filters contains both zeros and poles and the filter is recursive.
- IIR filters is not always stable. Hence, stability check is necessary.

Lecture summary

- To design an IIR filter, we always start with an analog filter, which serves as an analog prototype
- Then, the analog prototype is converted to a digital filter by Bilinear Transform (BLT) via the following substitution:

$$s = \frac{2}{T}\frac{1 - z^{-1}}{1 + z^{-1}}$$

Lecture summary

$$s = \frac{2}{T}\frac{1 - z^{-1}}{1 + z^{-1}}$$

- Problem with BLT:
  - The substitution is non-linear
  - This causes a non-linear relationship between analog frequencies and digital frequencies:

$$\Omega = \frac{2}{T}\tan\frac{\omega T}{2}$$

  - This is known as frequency warping
  - To remedy this effect, we pre-warp the analog frequency

Appendix – Bilinear Transform (BLT)

Here we show the derivation of the equation $\Omega = \frac{2}{T}\tan\frac{\omega T}{2}$ from the equation of BLT

- We denote the frequency of the analog filter by Ω and the frequency of the digital filter by ω

- The BLT under a sampling interval T is

$$s = \frac{2}{T}\frac{1 - z^{-1}}{1 + z^{-1}}$$

- **Relating *s* and *z* to Frequency:** In the context of frequency responses, the s-plane in the analog filter (where $s = j\Omega$ for steady-state sinusoidal inputs) can be related to the z-plane in the digital filter (where $z = e^{j\omega T}$) through the bilinear transform:

$$j\Omega = \frac{2}{T}\frac{1 - e^{-j\omega T}}{1 + e^{-j\omega T}}$$

Appendix – Bilinear Transform (BLT)

- Simplifying the Expression:

$$\frac{1-e^{-j\omega T}}{1+e^{-j\omega T}} = \frac{e^{j\omega T}-1}{e^{j\omega T}+1} = \frac{e^{\frac{j\omega T}{2}}\left(e^{\frac{j\omega T}{2}}-e^{-\frac{j\omega T}{2}}\right)}{e^{\frac{j\omega T}{2}}\left(e^{\frac{j\omega T}{2}}+e^{-\frac{j\omega T}{2}}\right)} = \frac{e^{\frac{j\omega T}{2}}-e^{-\frac{j\omega T}{2}}}{e^{\frac{j\omega T}{2}}+e^{-\frac{j\omega T}{2}}} = j\frac{\left(e^{\frac{j\omega T}{2}}-e^{-\frac{j\omega T}{2}}\right)/(2j)}{\left(e^{\frac{j\omega T}{2}}-e^{-\frac{j\omega T}{2}}\right)/2}$$

$$= j\frac{\sin\left(\frac{\omega T}{2}\right)}{\cos\left(\frac{\omega T}{2}\right)} = j\tan\left(\frac{\omega T}{2}\right)$$

- Therefore, the equation on last page becomes

$$j\Omega = \frac{2}{T}\frac{1-e^{-j\omega T}}{1+e^{-j\omega T}} = \frac{2}{T}j\tan\left(\frac{\omega T}{2}\right)$$

$$\Omega = \frac{2}{T}\tan\left(\frac{\omega T}{2}\right)$$

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